Icellmobilsoft wildfly dockerimage

Table of Contents

1. Common . . . 1
2. Dockerfile . . . 1
   2.1. wildfly/common . . . 2
      2.1.1. wildfly/microprofile . . . 4
   2.2. wildfly/oracle-jdbc . . . 6
   2.3. wildfly/postgresql-jdbc . . . 7
   2.4. wildfly/ibm-mq . . . 8
3. JMX usage . . . 9
4. Examples . . . 10
5. Release Notes . . . 10
   5.1. v2.0.0 . . . 10

1. Common

The repository is a collection of wildfly docker images used in icellmobilsoft, to keep projects and their builds on the same basis.

2. Dockerfile

The dockerfile structure is based on onion layering, can be reused to build another docker image. To do this, you need to give it the base image you want and it will add to it as needed. The required base images must meet certain conditions, which are usually bound by the names of the ENV parameters. These conditions are already described in Dockerhub Icellmobilsoft image repository.

LABEL

LABEL is a very important part of the builded docker images. They show all the data, that was an important factor in building the docker image. A sample of how this can be extracted from the docker image:

```
<CONSOLE>:~$ docker inspect icellmobilsoft/wildfly-full/oracle-jdbc-ibm-mq:1.0.0-
SNAPSHOT | jq -r '.[0].Config.Labels' ①
{
  "apt-get.install": "fontconfig,fonts-dejavu", ②
  "base.image": "icellmobilsoft/wildfly-full/oracle-jdbc:1.0.0-SNAPSHOT", ③
  "builder.images": "icellmobilsoft.hu/icellmobilsoft/builder-java17-galleon:1.1.0-
SNAPSHOT", ④
  "ee.configs": "managed-executor",
  "galleon.image": "icellmobilsoft/icellmobilsoft/builder-java17-galleon:1.1.0-
SNAPSHOT", ⑤
```

```
  "galleon.layers": "full", ⑥
  "ibm-mq.version": "com.ibm.mq:wmq.jmsra:9.1.0.8",
  "image.usage": "release", ⑦
  "java.image": "eclipse-temurin:17.0.7_7-jre", ⑧
  "logstash.version": "biz.paluch.logging:logstash-gelf:1.14.1",
  "microprofile.configs": "opentrace,metrics",
  "oracle-jdbc.version":
"com.oracle.database.jdbc:ojdbc10:19.7.0.0,com.oracle.database.nls:orai18n:19.7.0.0",
  "wildfly.modules": "logstash,oracle-jdbc,ibm-mq", ⑨
  "wildfly.properties.file": "/home/icellmobilsoft/config/wildfly.properties", ⑩
  "wildfly.version": "27.0.1.Final" ⑪
}
```

① the command itself that cuts the label part from the complete docker inspect response

② programs installed on linux base image

③ base image from which this particular image is derived (onion layering)

④ the images used in the build

⑤ name of the image that did the wildfly build

⑥ which galleon modules are used to build wildfly

⑦ image purpose. Intended for controlling, application image can only contain "release" value.

⑧ what image the java in the image comes from

⑨ what modules are manually installed in wildfly

⑩ the properties file used to start wildfly

⑪ wildfly version

Some of the labels are from base image, when creating them, you have to be careful when to overwrite them.

2.1. wildfly/common

In terms of layers, this is the only layer where the galleon is used to put together the wildfly application server. The base images are complemented with the logger java module serving logstash. For more information on layers, see the documentation at wildfly_galleon_layers.

In this case we will create these images from it:

- **icellmobilsoft/wildfly-default-ms:${VERSION}** - Image meets basic microservice requirements
- **icellmobilsoft/wildfly-default-jpa:${VERSION}** - Add jpa layer to the microservice image above
- **icellmobilsoft/wildfly-default-full:${VERSION}** - Full wildfly with all existing modules

The dockerfile is responsible for EXPOSE of the basic 8080 and 9990 (+ others) ports, and the wildfly startup CMD.

Table 1. Configuration options:

ENV kulcs	wildfly.properties kulcs	Default	Leírás
LOGGING_ROOT_LOGGER_LOGLEVEL	logging.root.logger.loglevel	INFO	Global log level
LOGGING_HU_ICELLMOBILSOFT_LOGGER_LOGLEVEL	logging.hu.icellmobilsoft.logger.loglevel	ALL	Global log level for "hu.icellmobilsoft" packages
LOGGING_COFFEE_ETCD_LOGGER_LOGLEVEL	logging.coffee.etcd.logger.loglevel	INFO	Log level regarding ETCD classes
LOGGING_K8S_NAMESPACE	-	-	Value of key `k8s_namespace` in the logged string (log pattern)
CONSOLE_LOGGING_ENABLED	console.logging.enabled	true	Enable plaintext console logging
CONSOLE_LOGGING_JSON_ENABLED	console.logging.json.enabled	false	Enable json console logging.
CONSOLE_LOGGING_JSON_FORMATTER_ZONEID	console.logging.json.formatter.zoneid	UTC	Default zoneid value for formatter datetime pattern
FILE_LOGGING_ENABLED	file.logging.enabled	false	Enable file logging. The maximum file size is 50MB by default.
LOGSTASH_ENABLED	logstash.enabled	false	Enable sending log entries to logstash server
LOGSTASH_BLOCKING_MODE	logstash.blocking.mode	DISCARD	Enable (BLOCK) or disable (DISCARD) blocking mode in case logs cannot be written
LOGSTASH_QUEUE_LENGTH	logstash.queue.length	50000	Size of the async queue length
LOGSTASH_HOST	logstash.host	127.0.0.1	Logstash server host.
LOGSTASH_PORT	logstash.port	8000	Logstash server port.
LOGSTASH_MODULE_ID	-	unknown	Set the logstash logging additionalFields `moduleId` key value.
LOGSTASH_MODULE_VERSION	-	unknown	Set the ogstash logging additionalFields `moduleVersion` key value.

ENV kulcs	wildfly.properties kulcs	Default	Leírás
LOGSTASH_K8S_NAMESPACE	-	unknown	Set the logstash logging additionalFields `K8S_NAMESPACE` key value.
LOGSTASH_MDC_FIELDS	-	-	Set the logstash logging additionalFields additional key values.

NOTE The value of the key defined in `wildfly.properties` has been overriden by the value of the key defined in `ENV`

2.1.1. wildfly/microprofile

Expands the base image created by wildfly/common with microprofile, executor and other config. This is necessary because wildfly builds the standalone.xml differently depending on the galleon layer. This layer combines all mp and ee configs that are good for full and ms wildfly.

In this case we create these images from it:

- **icellmobilsoft/wildfly-mp-ms:${VERSION}** - Basic microservice + microprofile configs
- **icellmobilsoft/wildfly-mp-jpa:${VERSION}** - Add jpa layer to the microservice image above
- **icellmobilsoft/wildfly-mp-full:${VERSION}** - Full wildfly with all existing modules + microprofile configs

Configuration options are inherited from the parent wildfly/common image and supplemented with the following:

Table 2. Configuration options:

ENV kulcs	wildfly.properties kulcs	Default	Leírás
MANAGED_EXECUTOR_SERVICE_CORE_THREADS	managed-executor-service.core-thread	16	Maximum number of Wildfly managed executors. (If it was not configured, t would be dynamic - [number of cpu cores] x 2)
OPENTRACING_JAEGER_AGENT_HOST	opentracing.jaeger.agent.host	localhost	Jaeger agent host
OPENTRACING_JAEGER_AGENT_PORT	opentracing.jaeger.agent.port	6831	Jaeger agent port

ENV kulcs	wildfly.properties kulcs	Default	Leírás
OPENTRACING_JAEGER_AGENT_SAMPLER_PORT	opentracing.jaeger.agent.sampler.port	5778	Jaeger agent sampler port
OPENTRACING_JAEGER_PROPAGATION	opentracing.jaeger.propagation	B3	Opentracing propagation (uber-trace vs B3)
METRICS_PREFIX	wildfly.metrics.prefix	wildfly	Default prefix for custom (microprofile-metrics-smallrye) metrics
HEALTH_EMPTY_LIVENESS_CHECKS_STATUS	-	UP	If the attribute is UP (by default), the server can be live when there are no liveness checks in the deployments. Setting the empty-liveness-checks-status attribute to DOWN will make this procedure fail when there are no liveness checks in the deployments.
HEALTH_EMPTY_READINESS_CHECKS_STATUS	-	UP	If the attribute is UP (by default), the server can be ready when there are no readiness checks in the deployments. Setting the empty-readiness-checks-status attribute to DOWN will make this procedure fail when there are no readiness checks in the deployments.

ENV kulcs	wildfly.properties kulcs	Default	Leírás
HEALTH_EMPTY_STARTUP_CHECKS_STATUS	-	UP	If the attribute is UP (by default), the server can be ready when there are no startup checks in the deployments. Setting the empty-startup-checks-status attribute to DOWN will make this procedure fail when there are no readiness checks in the deployments.
UNDERTOW_HTTP_MAX_POST_SIZE	undertow.http.max-post-size	10485760 (10MB)	Size limit of HTTP POST requests. Larger sizes are automatically dropped by Wildfly and returned with an error message.

NOTE The value of the key defined in `wildfly.properties` has been overriden by the value of the key defined in `ENV`

2.2. wildfly/oracle-jdbc

Expands the common image created by wildfly/microprofile with oracle feature.

In this case, these images are created from it:

- **icellmobilsoft/wildfly-mp-jpa-oracle-jdbc:${VERSION}** - `icellmobilsoft/wildfly-mp-jpa:${VERSION}` + oracle driver
- **icellmobilsoft/wildfly-mp-jpa-full-oracle-jdbc:${VERSION}** - `icellmobilsoft/wildfly-mp-full:${VERSION}` + oracle driver

Configuration options are inherited from the parent wildfly/microprofile image and supplemented with the following:

Table 3. Configuration options:

ENV kulcs	wildfly.properties kulcs	Default	Leírás
SPEC_DESCRIPTOR_PROPERTY_REPLACEMENT	spec-descriptor.property-replacement	true	Use variables in the property/descriptor files (ejb-jar.xml, persistence.xml)

ENV kulcs	wildfly.properties kulcs	Default	Leírás
ORACLE_DS_SERVICENAME	oracle.ds.servicename	icellmobilsoft	Service name for datasource, postfixed with DS automatically → icellmobilsoftDS by default
ORACLE_DS_CONNECTION_URL	oracle.ds.connection.url	-	Oracle DB connection url
ORACLE_DS_USERNAME	oracle.ds.username	-	Oracle DB connection user
ORACLE_DS_PASSWORD	oracle.ds.password	-	Oracle DB connection password
ORACLE_DS_NEW_CONNECTION_SQL	oracle.ds.new-connection-sql	BEGIN NULL; END;	Oracle DB new connection sql
ORACLE_DS_MIN_POOL_SIZE	oracle.ds.min-pool-size	1	Oracle DB connection min pool size
ORACLE_DS_MAX_POOL_SIZE	oracle.ds.max-pool-size	60	Oracle DB connection max pool size
DEFAULT_PERSISTENCE_HIBERNATE_DIALECT	-	org.hibernate.dialect.OracleDialect	Implementation constant value for usage in persistence.xml file

NOTE The value of the key defined in `wildfly.properties` has been overriden by the value of the key defined in `ENV`

Example for DEFAULT_PERSISTENCE_HIBERNATE_DIALECT usage: persistence.xml

2.3. wildfly/postgresql-jdbc

Expands the common image created by wildfly/microprofile with postgresql feature.

In this case, these images are created from it:

- **icellmobilsoft/wildfly-mp-jpa-postgresql-jdbc:${VERSION}** - `icellmobilsoft/wildfly-mp-jpa:${VERSION}` + postgresql driver
- **icellmobilsoft/wildfly-mp-full-postgresql-jdbc:${VERSION}** - `icellmobilsoft/wildfly-mp-full:${VERSION}` + postgresql driver

Configuration options are inherited from the parent wildfly/microprofile image and supplemented with the following:

Table 4. Configuration options:

ENV kulcs	wildfly.properties kulcs	Default	Leírás
SPEC_DESCRIPTOR_PROPERTY_REPLACEMENT	spec-descriptor.property-replacement	true	Use variables in the property/descriptor files (ejb-jar.xml, persistence.xml)
POSTGRESQL_DS_SERVICENAME	postgresql.ds.servicename	icellmobilsoft	Service name for datasource, postfixed with DS automatically → icellmobilsoftDS by default
POSTGRESQL_DS_CONNECTION_URL	postgresql.ds.connection.url	-	PostgreSQL DB connection url
POSTGRESQL_DS_USERNAME	postgresql.ds.username	-	PostgreSQL DB connection user
POSTGRESQL_DS_PASSWORD	postgresql.ds.password	-	PostgreSQL DB connection password
POSTGRESQL_DS_NEW_CONNECTION_SQL	postgresql.ds.new-connection-sql	BEGIN;END;	PostgreSQL DB new connection sql
POSTGRESQL_DS_MIN_POOL_SIZE	postgresql.ds.min-pool-size	1	PostgreSQL DB connection min pool size
POSTGRESQL_DS_MAX_POOL_SIZE	postgresql.ds.max-pool-size	60	PostgreSQL DB connection max pool size
DEFAULT_PERSISTENCE_HIBERNATE_DIALECT	-	org.hibernate.dialect.PostgreSQLDialect	Implementation constant value for usage in persistence.xml file

NOTE The value of the key defined in `wildfly.properties` has been overriden by the value of the key defined in `ENV`

Example for DEFAULT_PERSISTENCE_HIBERNATE_DIALECT usage: persistence.xml

2.4. wildfly/ibm-mq

Expands the common image created by wildfly/oracle-jdbc with ibm-mq feature.

In this case, these images are created from it:

- **icellmobilsoft/wildfly-mp-full-oracle-jdbc-ibm-mq:${VERSION}** - `icellmobilsoft/wildfly-mp-full-oracle-jdbc:${VERSION}` + ibm-mq driver

3. JMX usage

The modules for the wildfly jmx subsystem will be included in the images created by galleon, but by default the subsystem is disabled. On developer images it can be activated with a cli script, so there is no need to run a separate shell script for jmx.

configure-jmx.cli

```
/subsystem=jmx:add
/subsystem=jmx/remoting-connector=jmx:add(use-management-endpoint=true)

# management auth is turned off - not important for local execution anyway
/core-service=management/management-interface=http-interface:undefine-
attribute(name=http-authentication-factory)
/core-service=management/management-interface=http-interface:map-remove(name=http-
upgrade, key=sasl-authentication-factory)
```

To connect, you may need wildfly-cli.jar on the client classpath. This can be done for example as follows:

- download client jar

  ```
  mvn org.apache.maven.plugins:maven-dependency-plugin:2.8:get
  -Dartifact=org.wildfly.core:wildfly-cli:19.0.1.Final:jar:client
  ```

- add the downloaded jar to the classpath when starting the jmx client

  visualvm sample

  ```
  visualvm --cp:a ${HOME}/.m2/repository/org/wildfly/core/wildfly-
  cli/19.0.1.Final/wildfly-cli-19.0.1.Final-client.jar
  ```

  jconsole sample

  ```
  jconsole -J
  -Djava.class.path="${JAVA_HOME}/lib/jconsole.jar:${JAVA_HOME}/lib/tools.jar:${HOME}
  /.m2/repository/org/wildfly/core/wildfly-cli/19.0.1.Final/wildfly-cli-19.0.1.Final-
  client.jar"
  ```

  IMPORTANT

  for the above commands, you should make sure that the environment variables used exist, or replace them with absolute paths, `~` cannot be used for user home!

Wildfly allows jmx connection on the management port, url must be in the format `service:jmx:remote+http://localhost:${management.port}`.

4. Examples

persistence.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<persistence xmlns="https://jakarta.ee/xml/ns/persistence"
        xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
        xsi:schemaLocation="https://jakarta.ee/xml/ns/persistence
          https://jakarta.ee/xml/ns/persistence/persistence_3_0.xsd"
        version="3.0">

    <persistence-unit name="defaultPU" transaction-type="JTA">
        ...
        <properties>
            ...
            <property name="hibernate.dialect"
value="${env.DEFAULT_PERSISTENCE_HIBERNATE_DIALECT}" />
        </properties>
    </persistence-unit>
</persistence>
```

5. Release Notes

5.1. v2.0.0

The project has moved out of the internal repository, not counting updates, it is 1:1 compatible with version 1.1.0.

Changes compared to the internal version

- Renames:
  - wildfly-ms/default → icellmobilsoft/wildfly-default-ms
  - wildfly-jpa/default → icellmobilsoft/wildfly-default-jpa
  - wildfly-full/default → icellmobilsoft/wildfly-default-full
  - wildfly-ms/mp → icellmobilsoft/wildfly-mp-ms
  - wildfly-ms/jpa/mp → icellmobilsoft/wildfly-mp-jpa
  - wildfly-full/mp → icellmobilsoft/wildfly-mp-full
  - wildfly-ms/logstash → icellmobilsoft/wildfly-mp-ms
  - wildfly-ms/jpa-logstash → icellmobilsoft/wildfly-mp-jpa
  - wildfly-full/logstash → icellmobilsoft/wildfly-mp-full
  - wildfly-ms/oracle-jdbc → icellmobilsoft/wildfly-mp-jpa-oracle-jdbc
  - wildfly-full/oracle-jdbc → icellmobilsoft/wildfly-mp-full-oracle-jdbc
  - wildfly-ms/postgresql-jdbc → icellmobilsoft/wildfly-mp-jpa-postgresql-jdbc

- wildfly-full/postgresql-jdbc → icellmobilsoft/wildfly-mp-full-postgresql-jdbc
- wildfly-full/oracle-jdbc-ibm-mq → icellmobilsoft/wildfly-mp-full-oracle-jdbc-ibm-mq
- (replaced) wildfly-ms/gcp → icellmobilsoft/wildfly-mp-ms
- (replaced) wildfly-ms/jpa-gcp → icellmobilsoft/wildfly-mp-jpa
- (replaced) wildfly-full/gcp → icellmobilsoft/wildfly-mp-full
- (replaced) wildfly-ms/postgresql-jdbc-gcp → icellmobilsoft/wildfly-mp-jpa-postgresql-jdbc
- (replaced) wildfly-full/postgresql-jdbc-gcp → icellmobilsoft/wildfly-mp-full-postgresql-jdbc

- Environment changes:
  - (new with old value) `ORACLE_DS_SERVICENAME` default `icellmobilsoft+DS`
  - (new with old value) `POSTGRESQL_DS_CONNECTION_URL` default `icellmobilsoft+DS`
  - (new) `CONSOLE_LOGGING_JSON_ENABLED` default `false`
  - (new) `CONSOLE_LOGGING_JSON_FORMATTER_ZONEID` default `UTC`
  - (change value) `FILE_LOGGING_ENABLED` default `false` (changed from `true`)
  - (new with old value) `LOGSTASH_BLOCKING_MODE` default `DISCARD`
  - (new with old value) `LOGSTASH_QUEUE_LENGTH` default `50000`
  - (change value) `LOGSTASH_ENABLED` default `false` (changed from `true`)
- Debugging: The <code>--debug *:8787</code> option doesn’t need anymore in the <code>CMD exec</code> which you can ignore in the local Dockerfile mostly. The debug mode can be controlled by the <code>DEBUG</code> environment variable which is false by default. The default port is <code>*:8787</code> but you can override it with the <code>DEBUG_PORT</code> environment variable.

WARNING	Don't release docker images with DEBUG enabled into production environments!

What's new

- Baseimage version update to 1.4.0
- DEFAULT_PERSISTENCE_HIBERNATE_DIALECT variable added to the jdbc images to use in the local Dockerfiles:
  - wildfly-mp-*-oracle-jdbc image = `org.hibernate.dialect.OracleDialect`
  - wildfly-mp-*-postgresql-jdbc image = `org.hibernate.dialect.PostgreSQLDialect`

NOTE	*project-local.Dockerfile snippet* You can override default dialect with `PERSISTENCE_HIBERNATE_DIALECT` environment variable but you dont need to set it in the `docker-compose.yml` file mostly. ARG PERSISTENCE_HIBERNATE_DIALECT=$DEFAULT_PERSISTENCE_HIBERNATE_DIALECT ENV HIBERNATE_DIALECT=$PERSISTENCE_HIBERNATE_DIALECT